Syed Nabil Shah

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Education

University of Leicester Jun 2026

BSc Software Engineering - Achieved a First Class Honours (80%, top 5% of cohort)

- **Dissertation** on Cooperative AI Systems (MARL & Emergent Behaviour); preparing workshop submission.
- **Published preprint** on self-learning AI (Zenodo, 2025).

Experience

Minexx Jul 2024 - Jan 2026

Software Engineer Intern

- Engineered async microservices for large-scale news scraping, translation & topic modeling; across multilingual sources (**3K+ articles/day**, **95% dedup rate**, **+20% sentiment accuracy**).
- Developed **Tin Sentiment Bot**, a production NLP system combining GDELT data ingestion, multilingual processing & PostgreSQL warehousing to issue real-time buy/sell/hold signals for the tin market.
- Built fault-tolerant **ETL** pipelines enabling **24/7 automated** data sync on Compute Engine VM.
- Migrated company AI systems to scalable, production-grade GCP architecture, improving reliability for a platform handling **1M+ requests/day**.
- **Tech:** Python, Asyncio, Pandas, PostgreSQL, Firebase, GCP, NLTK, JavaScript, Grok API

Projects

Neuroplastic Collective Intelligence Ongoing

- Investigating whether neuroplastic communication mechanisms inspired by computational neuroscience can improve emergent coordination and functional specialisation in collective AI systems.
- Developing a multi-agent framework with adaptive graph-based communication and Hebbian-inspired plasticity.
- Exploring dynamic interaction topologies and information flow during cooperative learning.
- Evaluating emergent behaviour using graph/information-theoretic measures across cooperative benchmarks.
- **Tech:** Python, PyTorch, PettingZoo, NetworkX, NumPy

BidNet – Compute-Aware Neural Architecture via Differentiable Bidding 2026

- Built modular neural network where experts bid for compute based on predicted usefulness.
- Formulated routing using a **Lagrangian-style relaxation** to balance accuracy and compute.
- Implemented differentiable auction-style per-sample routing.
- Reduced compute via expert activation by ∼**3x** with comparable CIFAR-10 accuracy vs dense baseline.
- Improved accuracy & efficiency trade-off by ∼**8%** vs static top-k routing.
- Observed ∼**60% activation sparsity** with emergent expert specialisation.
- **Tech:** Python, PyTorch (MPS), NumPy

FianchettoX – Explainable Chess Engine 2025

- Built **2K+ LOC** C++ chess engine using **bitboards, alpha-beta pruning, and transposition tables** achieving <**150 ms** 6–10 ply search.
- Integrated neural evaluation via **ONNX** (**20 ms inference**), deployed across **3 Dockerised microservices**.
- **Tech:** C++, PyTorch/ONNX, Stockfish API, Next.js, Docker

Technical Skills

Languages: Python, Java, C++, JavaScript/TypeScript, SQL, Bash
AI/ML: TensorFlow, PyTorch, Scikit-learn, Hugging Face, Pandas, NumPy, spaCy, NLTK, BQML
Cloud & Data: GCP, AWS, Firebase, PostgreSQL
Frameworks & Tools: Next.js, Docker, Redis, Kafka, Git

Additional Achievements

Competitions: 1st place at IBM/Capital One Hackathon; Finalist & Prize winner at multiple Hackathons. Recently competed at Imperial AI Ventures Hackathon, Kaggle - Google ML & AI Mathematical Olympiad.
President, Founders Society: Launched the university's first startup accelerator, leading 100+ participants.
Debating: Top 30 worldwide (Oxford Schools'); 4× Regional Champion; 3× National Finalist.
Languages: Native English; Business-level German & Urdu.